LEARNING SESSION S6

Machine Learning: Basic to Advanced (for non-tech learners)

What ML is, every major algorithm in plain English, how models are judged, and how they reach production

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

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https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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🤖 Machine Learning: Basic to Advanced (for non-tech learners)

What ML is, every major algorithm in plain English, how models are judged, and how they reach production

Learning session · no maths background needed · links to the illustrated ML Class 1–7 notes for depth

Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#s6**

What ML is

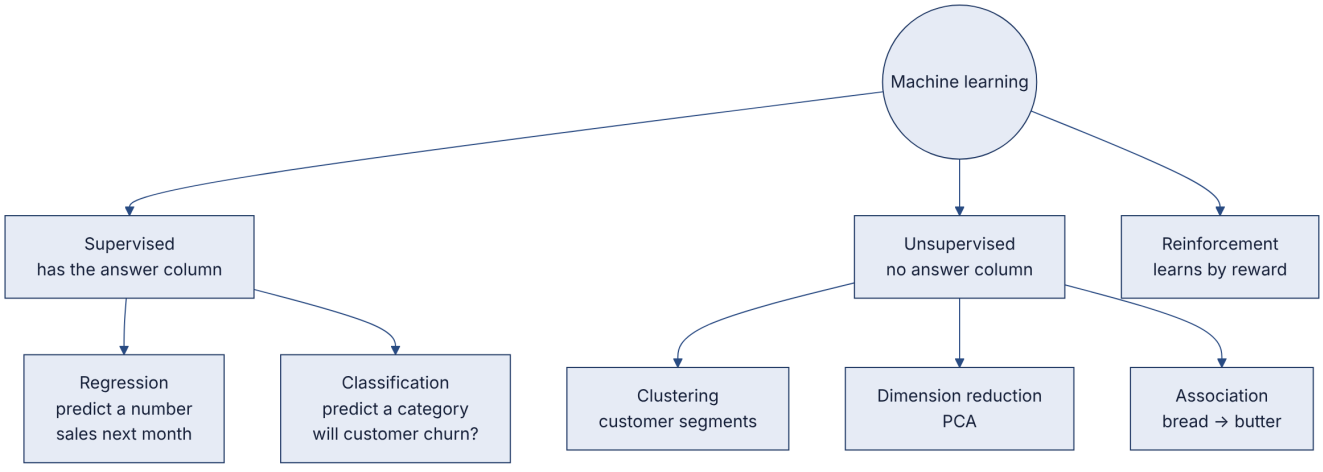
Machine learning is teaching a computer by **examples** instead of rules. Show it thousands of past loan applications labelled "repaid" or "defaulted", and it learns the pattern well enough to judge a new application.

Traditional rule

IF income < 3 lakh AND existing loans > 2 THEN reject. A person writes and maintains every rule.

Machine learning

Feed 50,000 past applications with outcomes. The algorithm finds the combinations of income, debt, age and history that predict default, and outputs a probability.



The three families of machine learning and their main tasks.

Everyday examples

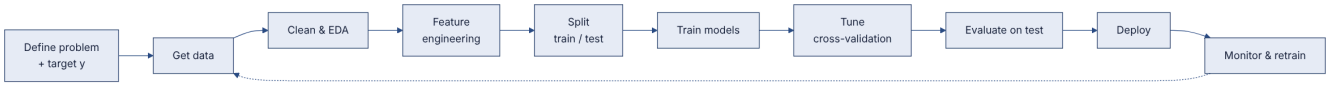
Where you see it	Task	Type
Gmail spam folder	Spam or not	Classification

Where you see it	Task	Type
Zomato delivery time estimate	Minutes to deliver	Regression
Netflix / Spotify recommendations	What you'll like next	Recommendation
Bank fraud SMS	Unusual transaction	Anomaly detection
Customer segments in marketing	Group similar customers	Clustering
Google Maps traffic, UPI demand planning	Future values	Forecasting
Review sentiment, ChatGPT	Understand/generate text	NLP / deep learning

Key words

Features (X)	Inputs: income, age, number of orders
Target / label (y)	What to predict: default yes/no, next month's sales
Training	The algorithm adjusts itself to fit past examples
Model	The learned pattern, saved and reused
Prediction / inference	Applying the model to new data
Train / test split	Learn on 70–80%, check on the unseen 20–30%

The ML workflow



Every ML project follows this loop. Most time goes into the first four boxes.

Step	Plain English	Example: predicting churn for a telecom
Define	What decision will the prediction support?	Who to call with a retention offer this month
Target	Exact definition of the outcome	Churn = ports out within 60 days
Data	Gather history with known outcomes	12 months of usage, bills, complaints, plan
Clean & explore	Fix gaps, errors, outliers; look at patterns	Missing tenure, duplicate accounts
Features	Turn raw data into useful signals	Drop in data usage, complaints last 30 days, days since recharge
Train & tune	Try models; tune settings with cross-validation	Logistic regression vs random forest vs XGBoost

Step	Plain English	Example: predicting churn for a telecom
Evaluate	Test on unseen data with business metrics	Catches 75% of churners in the top 20% risk list
Deploy & monitor	Score customers regularly; watch accuracy	Weekly scores to the CRM; retrain quarterly

Deep dive with real data and charts: **ML Class 3**, **Class 4**, **Class 6**.

Algorithms explained

Algorithm	Plain-English idea	Good for	Watch out
Linear regression	Draws the best straight line through the points	Sales vs ad spend; price estimates; explaining drivers	Only straight-line relationships unless you add features
Logistic regression	Linear score squeezed into a 0–1 probability	Credit risk, churn; explainable (regulators like it)	Misses complex interactions
Decision tree	A flowchart of yes/no questions learnt from data	Easy-to-explain rules	One deep tree overfits
Random forest	Hundreds of trees on random samples vote	Strong all-rounder for tables	Less interpretable, bigger
Gradient boosting (XGBoost, LightGBM, CatBoost)	Trees built one after another, each fixing the previous errors	Best accuracy on most business tabular data	Needs tuning; can overfit
K-nearest neighbours	"You are like your closest neighbours"	Simple baselines, recommendations	Needs scaling; slow on big data
Naive Bayes	Bayes' rule assuming features independent	Spam and text classification	Independence assumption is rough
Support vector machine	Widest possible margin between classes	Smaller, high-dimensional data	Slow on large data; hard to explain
K-means clustering	Puts customers into K groups around centres	Segmentation	Choose K (elbow/silhouette); scale features
Hierarchical clustering	Builds a tree of nested groups (dendrogram)	Small data, exploring structure	Slow for big data
PCA	Compresses many correlated columns into a few	Visualising, removing redundancy	Components are hard to name
Association rules (Apriori)	"Bread → butter" with support, confidence, lift	Market basket, cross-sell	Many trivial rules

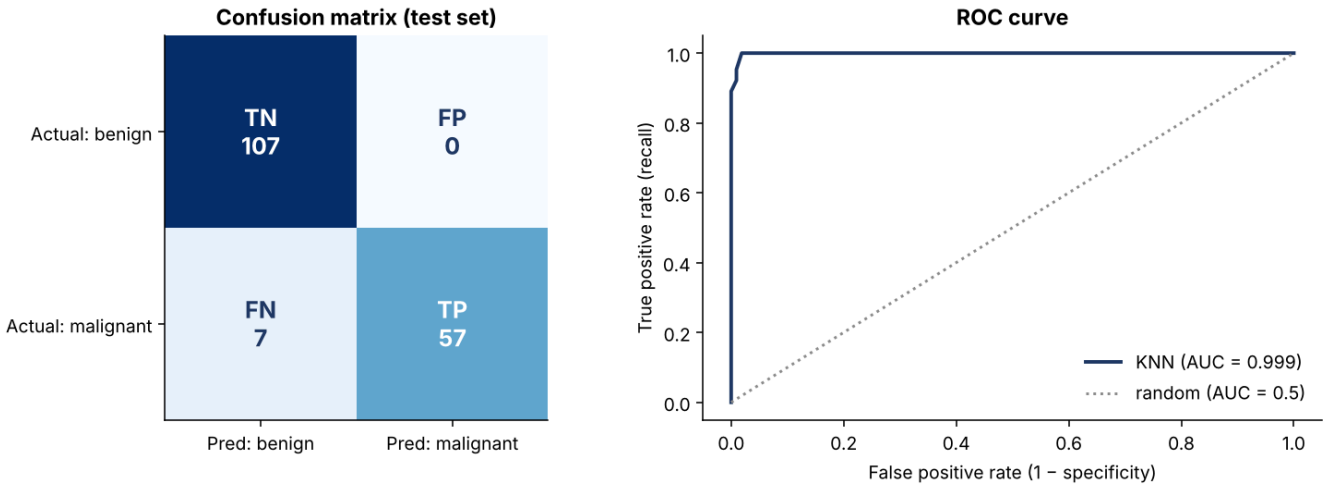
Algorithm	Plain-English idea	Good for	Watch out
Time-series (ARIMA, ETS, Prophet)	Uses trend + seasonality to forecast	Demand, cash flow, call volumes	Breaks when patterns shift
Neural networks / deep learning	Layers of simple units learning complex patterns	Images, speech, text, LLMs	Needs lots of data and compute

Is the model good?

Regression metrics

MAE	Average size of error: "off by ₹1,200 on average"
RMSE	Like MAE but punishes big misses more
MAPE	Average % error: "forecast is 8% off"
R^2	Share of variation explained (0 to 1)

Classification metrics



Confusion matrix and ROC curve (see ML Class 7 for the full worked example).

Accuracy	Share of correct predictions. Misleading if one class is rare.
Precision	Of those flagged, how many were right? (Avoid wasting offers/false alarms.)
Recall	Of the real positives, how many did we catch? (Don't miss fraud or disease.)
F1	Balance of precision and recall
ROC-AUC	How well the model ranks positives above negatives (0.5 random, 1.0 perfect)
Lift / gains	"Top 10% of scores contain 40% of churners" – the metric business teams love

Overfitting in one sentence

A model that memorises the training data (like a student memorising last year's paper) scores brilliantly in practice and fails the real exam. We prevent it with test sets, cross-validation, simpler models and more data. Picture: ML Class 6.

Advanced topics

Feature engineering	Ratios, recency, trends, date parts, text length; often matters more than the algorithm
Imbalanced data	Class weights, SMOTE oversampling, choosing the right threshold
Hyper-parameter tuning	GridSearchCV, RandomizedSearchCV, Bayesian search (Optuna)
Ensembles	Bagging, boosting, stacking several models
Explainability	Feature importance, SHAP values ("income lowered this applicant's risk by 8 points"), partial dependence
Text mining / NLP	Tokenise → clean (stop words, lemmatise) → bag-of-words / TF-IDF / embeddings → classify sentiment or topics
Recommenders	Collaborative filtering, matrix factorisation, embeddings
Deep learning	CNNs for images, RNN/LSTM and Transformers for sequences and text
AutoML / no-code	Azure AutoML, Google Vertex AI, DataRobot, H2O, PyCaret; Copilot or ChatGPT data analysis for quick models
MLOps	Versioning, deployment, monitoring and retraining (see ML Ops module)
Responsible AI	Fairness across groups, privacy (DPDP Act in India), transparency, human review of high-impact decisions

Use cases by industry

Industry	Use case	Typical model	Business metric
Banking	Credit scoring, fraud, collections priority	Logistic regression, gradient boosting	Default rate, fraud loss, recovery %
Retail / e-commerce	Demand forecasting, recommendations, churn, pricing	Time series, collaborative filtering, boosting	Stock-outs, conversion, AOV
Telecom	Churn prediction, network fault prediction	Boosting, survival models	Churn rate, retention ROI
Healthcare	Readmission risk, image diagnosis support	Logistic regression, CNNs	Readmissions, sensitivity
Operations / call centres	Call-volume forecasting, staffing, intent routing	ETS/ARIMA, text classification	Service level, AHT, cost per contact

Industry	Use case	Typical model	Business metric
HR	Attrition risk, resume screening (with bias checks)	Logistic regression, NLP	Attrition, time-to-hire
Marketing	Segmentation, propensity to buy, uplift modelling	K-means, boosting	Campaign ROI, CAC

Interview Q&A

Q Explain machine learning to a 10-year-old.

It's like learning to recognise dogs: after seeing many pictures labelled "dog" and "not dog", you can tell for a new picture. The computer does the same with data.

Q Supervised vs unsupervised?

Supervised learns from examples with known answers (predict churn). Unsupervised finds structure without answers (group customers).

Q What is overfitting and how do you prevent it?

Memorising noise in training data so the model fails on new data. Prevent with train/test splits, cross-validation, simpler models, regularisation and more data.

Q Precision vs recall with a business example?

Fraud: recall = share of frauds caught; precision = share of alerts that were real fraud. Missing fraud is costly, so prioritise recall while keeping alerts manageable.

Q Why is random forest better than one decision tree?

Averaging many trees trained on different samples reduces variance, so predictions are more stable and accurate.

Q How do you explain a model's prediction to a manager?

Show the top drivers (feature importance or SHAP), a few example customers, and the business result (e.g. calls saved, revenue protected).

Resources

- ▶ YouTube: Simplilearn machine learning full course · ▶ YouTube: Simplilearn machine learning tutorial for beginners
- ▶ YouTube: StatQuest machine learning playlist · ▶ YouTube: Andrew Ng machine learning specialization
- ▶ YouTube: codebasics machine learning tutorial python · ▶ YouTube: Krish Naik complete machine learning playlist
- 📘 Google Machine Learning Crash Course · 📘 Kaggle Learn: Intro to ML
- 📘 scikit-learn: choosing the right estimator